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Paediatric Strategy Forum for medicinal product development of chimeric antigen receptor T-cells in children and adolescents with cancer ACCELERATE in collaboration with the European Medicines Agency with participation of the Food and Drug Administration²

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Abstract

The seventh multi-stakeholder Paediatric Strategy Forum focused on chimeric antigen receptor (CAR) T-cells for children and adolescents with cancer. The development of CAR T-cells for patients with haematological malignancies, especially B-cell precursor acute lymphoblastic leukaemia (BCP-ALL), has been spectacular. However, currently, there are scientific, clinical and logistical challenges for use of CAR T-cells in BCP-ALL and other paediatric malignancies, particularly in acute myeloid leukaemia (AML), lymphomas and solid tumours. The aims of the Forum were to summarise the current landscape of CAR T-cell therapy development in paediatrics, to identify current challenges and future directions, with consideration of other immune effector modalities and ascertain the best strategies to accelerate their development and availability to children.

Although the effect is of limited duration in about half of the patients, anti-CD19 CAR T-cells produce high response rates in relapsed/refractory BCP-ALL and this has highlighted previously unknown mechanisms of relapse. CAR T-cell treatment as first- or second-line therapy could also potentially benefit patients whose disease has high-risk features associated with relapse and failure of conventional therapies. Identifying patients with very early and early relapse in whom CAR T-cell therapy may replace haematopoietic stem cell transplantation and be definitive therapy versus those in whom it provides a more effective bridge to haematopoietic stem cell transplantation is a very high priority. Development of approaches to improve persistence, either by improving T cell fitness or using more humanised/fully humanised products and co-targeting of multiple antigens to prevent antigen escape, could potentially further optimise therapy.

Many differences exist between paediatric B-cell non-Hodgkin lymphomas (B-NHL) and BCP-ALL. In view of the very small patient numbers with relapsed lymphoma, careful prioritisation is

²The participants are given in Appendix section.

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needed to evaluate CAR T-cells in children with Burkitt lymphoma, primary mediastinal B cell lymphoma and other NHL subtypes. Combination trials of alternative targets to CD19 (CD20 or CD22) should also be explored as a priority to improve efficacy in this population. Development of CD30 CAR T-cell immunotherapy strategies in patients with relapsed/refractory Hodgkin lymphoma will likely be most efficiently accomplished by joint paediatric and adult trials.

CAR T-cell approaches are early in development for AML and T-ALL, given the unique challenges of successful immunotherapy actualisation in these diseases. At this time, CD33 and CD123 appear to be the most universal targets in AML and CD7 in T-ALL. The results of ongoing or planned first-in-human studies are required to facilitate further understanding. There are promising early results in solid tumours, particularly with GD2 targeting cell therapies in neuroblastoma and central nervous system gliomas that represent significant unmet clinical needs. Further understanding of biology is critical to success.

The comparative benefits of autologous versus allogeneic CAR T-cells, T-cells engineered with T cell receptors T-cells engineered with T cell receptor fusion constructs, CAR Natural Killer (NK)-cell products, bispecific T-cell engager antibodies and antibody-drug conjugates require evaluation in paediatric malignancies.

Early and proactive academia and multi-company engagement are mandatory to advance cellular immunotherapies in paediatric oncology. Regulatory advice should be sought very early in the design and preparation of clinical trials of innovative medicines, for which regulatory approval may ultimately be sought. Aligning strategic, scientific, regulatory, health technology and funding requirements from the inception of a clinical trial is especially important as these are very expensive therapies.

The model for drug development for cell therapy in paediatric oncology could also involve a 'later stage handoff' to industry after early development in academic hands. Finally, and very importantly, strategies must evolve to ensure appropriate ease of access for children who need and could potentially benefit from these therapies.

Keywords

Paediatric oncology; CAR T-cell; Adoptive cellular immunotherapy; Paediatric Strategy Forum; Drug development; Cancer therapeutics

1. Introduction

The goal of a multi-stakeholder Paediatric Strategy Forum¹ [2–5] is to share information between all stakeholders in a pre-competitive setting to inform paediatric drug development strategies and subsequent decisions. This is achieved by providing a unique opportunity to facilitate dialogue and enable constructive interactions between all stakeholders (regulators, pharmaceutical companies, clinical academics and patient advocates) on topics requiring discussion in drug development in children and adolescents with malignancy. In this way, novel drugs with a similar mechanism of action can then be 'compared' in a non-competitive space, such that precious resources are not wasted, and paediatric patients are not enrolled on sub-optimal clinical studies. These Forums provided unprecedented opportunities for

meaningful interaction between all stakeholders on topics that might cause a feasibility problem from an industry or academic standpoint in paediatric or adolescent cancer drug development. They facilitate the development of and scientifically driven discussion about the best choices of innovative medicines for the treatment of children with cancer and ultimately accelerate the introduction of these medicines into the standard of care for children.

The development of chimeric antigen receptor (CAR) T-cell immunotherapy [6] for patients with cancer over the last 11 years has been dramatic and rapid [7–9]. CAR T-cells were first used to treat advanced non-Hodgkin lymphoma (NHL) in 2010, chronic lymphocytic leukaemia in 2011, paediatric and adolescent/young adult (AYA) B-cell precursor acute lymphoblastic leukaemia (BCP-ALL) in 2012 and most recently multiple myeloma. The first agency-approved CAR T-cell therapy was tisagenlecleucel for the treatment of paediatric/AYAs with relapsed/refractory BCP-ALL (approved by the FDA in 2017 and EMA in 2018). This was followed by the approvals by the EMA and FDA for tisagenlecleucel for adults with relapsed/refractory diffuse large B-cell lymphoma (DLBCL), axicabtagene ciloleucel for adults with relapsed/refractory large B-cell lymphoma or follicular lymphoma, brexucabtagene autoleucel for adults with relapsed/refractory mantle cell lymphoma, lisocabtagene maraleucel for adults with relapsed/refractory large B-cell lymphoma and idecabtagene vicleucel for adults with relapsed/refractory multiple myeloma. Over 1100 trials of CAR T-cells have now been completed or are ongoing in children and adults with high-risk cancers [10]. However, numerous biological, clinical and manufacturing challenges exist in the use of CAR T-cells in the treatment of cancer and specifically in paediatric malignancies [11]. To develop CAR T-cell therapy effectively and efficiently for the benefit of children with cancer, there is a need for close cooperation between academia, industry, regulators and patient advocates. It was, therefore, considered timely to hold a Paediatric Strategy Forum focussing on CAR T-cell therapies in children and adolescents with cancers. The aims of the Forum were to summarise the current landscape of CAR T-cell therapy development in paediatrics to identify current challenges and future directions, with consideration of other immune effector modalities, such as T-cell receptor (TCR)-engineered T-cells, TCR fusion construct T-cells (TRuC) and CAR-NK cell products, and ascertain the best strategies to accelerate their development and availability to children.

The meeting was organised by ACCELERATE in collaboration with the European Medicines Agency (EMA) with participation of the United States Food and Drug Administration (FDA) and was held virtually on 25th, 26th and 27th May 2021 with 226 participants: 95 international paediatric haematology–oncology and immunotherapy experts from Europe, US, Canada and Australia; 54 representatives from 13 pharmaceutical companies in Europe, US, China and Singapore (Autolus Limited, Celgene/BMS, Cellectis, CRISPR Therapeutics, Gracellbiotechnologies Inc, GlaxoSmithKline, Kite – a Gilead company, Miltenyi Biomedicine, Novartis, Syncopation Life Sciences, Takeda Pharmaceuticals, TCR [2] Therapeutics and Tessa Therapeutics); 13 patient advocates from Europe, US and Canada (representatives from Andrew McDonough B + Foundation, Ac2orn, Bone Cancer Research Trust, Children’s Cancer Cause, KIDS V CANCER, KickCancer, PORT (Paediatric Oncology Reference Team), Solving Kids’ Cancer, Solving Kids’ Cancer UK, World Duchenne Organization and European Patients Forum, Zoé4life

and Childhood Cancer International); 61 regulators from the EMA (including Paediatric Committee [PDCO] and Committee for Advanced Therapies [CAT]) and national competent authorities within the EU regulatory network, Health Technology Assessment (HTAs) bodies; US FDA; UK Medicines and Healthcare Products Regulatory Agency; Health Canada as observers, and three organisers. A comprehensive overview of the development of CAR T-cells in BCP-ALL, T-ALL, acute myeloid leukaemia (AML) and lymphomas was presented by academic experts, as well as emerging approaches for CAR T-cell strategies in solid tumours and remaining challenges. Details of thirteen CAR T-cell, TCR T-cell, TRuC T-cell and CAR NK-cell products that are or will be under evaluation specifically in paediatric trials were also highlighted by industry representatives. The Forum concluded with the patient advocate perspective and a multi-stakeholder strategic discussion.

2. CAR T-cells in BCP-ALL

Ten percent of children with BCP-ALL relapse following standard chemotherapy regimens and many relapses cannot be effectively rescued even with allogeneic haematopoietic stem cell transplantation (HSCT) [12,13]. BCP-ALL thus remains a major cause of death from childhood malignancy given its relatively common incidence. Furthermore, deaths from therapy, acute toxicity and long-term adverse events, especially in those patients who have received HSCT, need to be prevented [14]. The outcome of patients is particularly poor for those patients with BCP-ALL with (a) adverse genetic features at initial presentation; (b) high levels of minimal residual disease after initial induction and particularly after consolidation therapy; (c) early medullary relapse and (d) multiple relapses. CAR T-cell therapy could potentially benefit these patients for whom conventional therapies fail. A long-term vision is that CAR T-cell therapy could replace many components of toxic standard treatment and, in the future, substantially shorten the duration of therapy needed to cure children with BCP-ALL.

The ELIANA (Novartis) trial of the autologous anti-CD19 CAR T-cell product, tisagenlecleucel, in children and AYAs with relapsed/refractory BCP-cell ALL (with central manufacturing and global distribution to 25 sites across 11 countries) reported a complete response (CR)/complete response with incomplete cell count recovery (CRi) rate of 81% in patients who had T cells infused, with a probability of relapse-free survival (RFS) of 59% among responders at 12 months post-therapy. Seventy-nine of 92 patients enrolled had CAR T-cells infused [15,16], highlighting critical factors of cell manufacturing difficulties or leukaemia progression and clinical instability that precluded infusion. Although the response is of limited durability in about half of the patients, other academic paediatric and adult studies of autologous CD19 CAR T-cell therapy products have confirmed their high initial response rates [17–22]. Multiple studies have now shown that post-CD19 CAR T-cell relapse often occurs because of antigen escape with the loss of CD19 by acquired somatic gene mutations in the leukaemic clone [23]. Failure of expansion and persistence [24,25] of CAR T-cells is another mechanism of relapse. Recent studies have also reported that prior treatment with other CD19-targeted therapy (e.g. blinatumomab) [26–28] and a higher medullary tumour burden before CAR T-cell treatment are factors associated with higher relapse rates after CD19 CAR T-cell therapy in this population [25,29]. CAR T-cells containing costimulatory domains of CD28 (versus 4–1BB, as in tisagenlecleucel) have

also been shown to be effective at inducing remissions and can effectively be a bridge to subsequent HSCT, but do not persist *in vivo* beyond a few weeks and are unlikely to be curative as a stand-alone therapy in children with relapsed/refractory B BCP-ALL [17,18].

To improve CAR-T cell persistence and RFS, humanised [25] or fully human products and/or products with improved T cell fitness are under evaluation. Alternative targets in BCP-ALL have also shown promise, including CD22, CD20 [30,31] and CRLF2 [32,33]. A clinical trial in paediatric and AYA patients with relapsed/refractory disease of CD22 CAR T-cells showed robust remission induction, including those with CD19-negative relapse after CD19 CAR T-cells. This has facilitated subsequent HSCT in some patients [34–36]. It is currently thought that relapse via antigen escape mechanisms could be potentially prevented by co-targeting multiple antigens (e.g. dual CD19 × CD22 CAR T-cells) [37] either in bivalent or bicistronic constructs or by sequential or concomitant co-infusion of monovalent anti-CD19 and anti-CD22 CAR-T cells [17,22]. Multi-antigen targeting is under active pre-clinical and clinical investigation, including dual- and even triple-antigen targeting strategies [38]. To date, it has not yet been clearly demonstrated that a multi-antigen approach will improve RFS nor whether a simultaneous or sequential approach of co-targeting may be superior.

2.1. Role of CAR T-cells in BCP-cell ALL in front-line therapy

The therapeutic benefit of CAR T-cells in patients with newly-diagnosed National Cancer Institute high-risk BCP-ALL with persistent minimal residual disease after two cycles of chemotherapy [39] (induction and consolidation) is currently being investigated in the CASSIOPEIA (Novartis) trial, given the documented poor clinical outcomes of this population with 5-year event-free survival of 39% [40]. The primary outcome of this single-arm study of 140 patients is 5-year RFS compared with historical control patients treated with chemotherapy. To account for the capacity of CAR T-cells to replace HSCT in this population, in addition to the primary end-point of RFS, a secondary end-point of disease-free survival without allogeneic HSCT is being explored [41]. The trial is expected to report the first results in approximately 2025.

2.2. Role of CAR T-cells in B-cell ALL at very early/early relapse

The role of CAR T-cells in an early and very early relapse in a landscape of chemotherapy (+/-blinatumomab) and HSCT requires clarification. This is especially important after recent randomised trials established a clear benefit with substantially superior RFS after blinatumomab as a bridge to HSCT [12,13] in patients in first early relapse of BCP-ALL or late relapse with minimum residual disease (MRD) positivity after reinduction chemotherapy. Two strategies are currently being followed, some centres use CAR T-cell therapy as a bridge to HSCT, other centres use CAR T-cell therapy followed by careful monitoring of MRD reappearance and use HSCT only in those cases. The first strategy is focused on the prevention of relapses, and the second strategy is the prevention of HSCT with its late effects. Which one leads to better survival is unknown. Determining if CAR T-cell therapy is a bridge to HSCT or if it is curative and could replace HSCT in some patients is a very high priority. Ideally, this question should be assessed within a randomised study. It is to be decided if such a trial should target CD19 alone or use an advanced product

co-targeting CD19 and CD22, and which optimal product(s) should be investigated. Given the high priority of this research question, a partnership between academia and industry will be needed as it is unlikely this will be a solely industry-sponsored study due to timelines, cost and feasibility of enrolment. Furthermore, regulatory involvement will be required on trial design to fulfil potentially regulatory requirements and regulatory agreement on trial design. An academic-sponsored study within the framework of an international childhood leukaemia cooperative study group with industry support is the favoured option to answer this question efficiently and definitively. The design of any randomised study is not trivial as not only all patients would have a CAR T-cells product manufactured effectively but also because patients proceeding to allogeneic HSCT need to be in a CR, which is not the case for CAR T-cells.

3. CAR T-cells in AML

Approximately 30% of children with AML will die from their disease despite maximally intensive chemotherapy and HSCT, often in the first remission. Early relapse (<12 months from diagnosis) and any relapse after HSCT are associated with $\leq 20\%$ survival [42]. A key challenge is that there are no known AML-specific antigens that are uniquely expressed on the malignant clone, but not on normal myeloid precursors, as a basis for immunotherapy [43–45]. The best-studied, most validated and most selective immunotherapeutic targets currently for childhood AML are CD123 [46–54] and CD33 [55–58]. Hence, most of the CAR T-cell clinical development, to date, has focused on CD33 or CD123 targeting given that the targets are expressed in >85% of AML, and clinical safety data with several antibody-based immunotherapies are now available [59–61]. Other targets under active exploration include mesothelin [62], CD56 [63] (expressed by the important rare *CBFA2T3-GLIS2* fusion subgroup, occurring exclusively in the very young paediatric population with a high risk acute megakaryoblastic leukaemia subtype and associated with a dismal prognosis), FLT3, CD38, CD44v6 [64] and CLEC12A/CLL-1 [65]. Early phase studies with paediatric enrolment are ongoing with CAR T-cells targeting CD33, CD123, CD44v6, and CLL-1 [59–61,64,65]. The challenges with these trials are a heavily pretreated relapsed/refractory AML patient population, which has made autologous T-cell apheresis quite difficult, rapid AML progression while awaiting CAR T-cell manufacturing, lack of effective bridging chemotherapy options during CAR T-cell manufacturing, and difficulty in enrolling patients ≥ 16 years with relapsed/refractory AML at paediatric institutions. Because of the nature of the target antigens which are co-expressed on normal haematopoietic progenitor cells, current CAR T-cell therapy in AML may be associated with appreciable on-target/off-tumour toxicity. At this time, most AML CAR T-cell immunotherapy does not aim to provide long-term persistence of CAR T-cells but rather aims to provide effective remission reinduction and bridge to subsequent transplant which in itself may be important for chemo-refractory patients. Early results of clinical studies demonstrate that the safety of AML CAR T-cell immunotherapy has been reasonable without as much ‘bystander’ toxicity as initially feared with some deep responses that have been transplant-enabling.

3.1. Role of CAR T-cells in the therapy of AML

It remains unknown whether novel improved CAR T-cell designs will allow AML immunotherapy without subsequent HSCT in the future. Evaluation of the relative value of more readily-available bispecific T-cell engager antibodies and antibody-drug conjugates (ADCs) targeting myeloid antigens, which also provide preliminary antigen-redirected T cell responses as a bridge to HSCT is also of paramount importance. The potential efficacy of dual targeting CAR T-cells with two or more different AML antigens, the optimal immunotherapy combination with small molecule inhibitors and optimal indication and/or sequence of CAR T-cells and T-cell engagers requires clarification. Because persistence of CAR T-cells is currently not the aim for a bridge-to-transplant strategy, allogeneic CAR T-cells (including multiple administrations), allogeneic NK cells, CAR NK-cells and other adoptive cellular therapy may have specific roles in treating some patients with high-risk/relapsed AML. The results of ongoing or planned first-in-child studies are required to understand the role of CAR T-cell therapy in AML. Further engineered T cell therapies targeting AML-restricted antigens in 'boutique' subtypes with a high expression on leukaemic blasts but with low to no expression in normal haematopoiesis (e.g. mesothelin), warrants clinical investigation.

4. CAR T-cells in T-cell ALL

Immune-targeted therapy options in patients with relapsed/refractory T-ALL are limited because most targetable antigens are expressed on normal T cells [66]. This results in two challenges using CAR T-cells in T-cell ALL (i) fratricide of CAR T-cells resulting in impaired expansion during manufacturing; (ii) targeting normal T cells resulting in immunodeficiency, which in contrast to B-cell depletion cannot currently be compensated in patients. In addition, similar to BCP-cell ALL, subclones or mutants with low or lost antigen expression can escape and produce antigen-negative relapses [66]. There are two groups of T-ALL antigens that are currently being targeted with CAR T-cell therapies, (i) pan-T antigens (CD5, CD7), which are widely expressed in T-ALL and are therefore likely to allow benefit for most patients, however, fratricide and T-cell aplasia need to be overcome; and (ii) subset-restricted (CD1a, TRBC1) antigens with which fratricide/aplasia are easier to overcome, but the expression is restricted to smaller subgroups of T-ALL [66]. Clinical development of CAR T-cells for T-ALL to date has been most advanced for CD5 [67]. These clinical studies show that CD5-specific CAR T-cells can be manufactured despite fratricide. Because of down-regulation of CD5 on CAR T-cells during *in vitro* expansion, endogenous T cells are reduced but not eliminated post CD5 CAR T-cell infusion. Therapy is safe but responses are (currently) sub-optimal [67]. Evidence suggests that fratricide by CD7 CAR T-cells can be overcome by gene-editing because CD7 expression appears not to be a prerequisite for T cell function [66,68–70]. In addition, the adoptive transfer of gene-edited CD7-negative T-cells could also overcome *in vivo* T-cell depletion by CD7-specific CAR T-cells [67]. There are ongoing clinical studies of CD7-modified CAR T-cells demonstrating promising activity in relapsed/refractory T-ALL [71–75], including durable remissions in a subset of patients [75]. Investigating additional T-ALL targets and/or co-targeting strategies, allogeneic CAR T-cells or CAR NK-cells is also warranted. As with AML, the results of

ongoing or planned first-in-child studies are required to facilitate further understanding and clinical progress.

5. CAR T-cells and other adoptive cell therapies in solid tumours

Cellular immunotherapies are currently being evaluated in solid tumours by numerous academic research teams and industry with some promising early results recently reported. CAR T-cells targeting the non-protein ganglioside GD2 have been developed over many years [76–80]. Clinical activity in neuroblastoma has been recently reported, including complete remissions with no evidence of on-target neurotoxicity. This is consistent with a therapeutic window for on-target/ontumour targeting GD2 without deleterious effects on normal tissues, where GD2 can be present at low densities [78,81]. Pre-clinical studies of GD2-CAR T-cells have shown potent eradication of H3K27M diffuse midline glioma tumours in murine models [82]. CAR T-cells administered directly into the central nervous system in murine models were also found to be more potent than those administered intravenously [83,84]. An ongoing investigator-initiated Phase I clinical trial has reported manageable toxicity without on-target neurotoxicity, improved clinical symptoms and decreased tumour size in response to GD2-specific CAR T-cells, administered intravenously, in patients with diffuse midline glioma [85]. These responses are particularly notable as these tumours have one of the worst prognoses of all paediatric malignancies, with no known curative strategies despite decades of research and many trials. Innovative approaches aiming to improve the antitumour activity of GD2-specific CAR T-cells include engineering cytokine genes to allow optimal expansion (e.g. C7R-GD2 CAR T-cells) and the use of alternative effector cells, such as autologous NKT cells [86–90]. Glypican 3 (GPC3)-specific CAR T-cells have also shown some early response signals in patients with hepatocellular carcinoma [91] and are now being evaluated in combination with interleukin-15 [92] and –21 [93]. This work provides important feasibility for ongoing and planned studies of CAR T-cells targeting an alternative glypican, such as GPC2, in neuroblastoma [94]. In addition, HER2 CAR T-cells for glioblastoma and with or without PD1-blockade for paediatric sarcomas [95,96], L1CAM for neuroblastoma [97], EGFR806-specific CAR T-cells in glioblastoma [98] and B7H3-specific CAR T-cells [84,99] for various solid cancers of childhood are showing potential.

Overall, targetable antigens in paediatric solid cancers are rare and may be complicated by co-expression on normal tissues and/or heterogeneous expression in tumours. It is important to identify and prioritise the most promising antigens, which currently include GD2 and B7H393 [84] and NY-ESO-1 as a target for TCR. Novel CAR designs, for example, allowing T-cell activation only in the presence or absence of an additional marker, and combinations need to be investigated, and novel trial designs, such as ‘pick the winner, drop the loser,’ should be used. The challenges in developing CAR T-cells in solid tumours include the fact that targets lack either tumour specificity or homogeneous expression or both and that there are various mechanisms contributing to T-cell dysfunction in the tumour microenvironment, which makes it difficult for CAR T-cells to infiltrate the tumour, expand and persist. TCR engineering is an alternative approach that targets antigens both on the cell surface and within the cell but requires peptide presentation on MHC, which may be defective in cancer cells, and which restricts patient eligibility by human leukocyte antigens

(HLA) type. NY-ESO-1- and MAGEA4-specific T cells for the treatment of synovial sarcoma are a promising example currently evaluated in both adults and the paediatric population [100,101]. Engineered T cell platforms that leverage the power of the entire TCR complex (as opposed to just the CD3 zeta chain used in CAR-T constructs) while avoiding HLA restriction, so-called TRuC T-cells, may improve T cell trafficking and persistence and therefore should be investigated in paediatric solid cancers [102].

Generally, the development of CAR T-cell and other adoptive cell therapy products for adults is not a clearly defined route for meaningful paediatric studies. Because of the distinct biologies and cells of origin of paediatric and adult solid cancers, only a small overlap exists for antigens of interest. However, the inclusion of adolescents in adult programs, if scientifically justified (as with NY-ESO-1 specific T-cells), is strongly encouraged. There are various potential models for development (i) development in children first by academia then to industry; (ii) joint development by academia and industry; (iii) standalone academic development within a business model.

In summary, early clinical trials with CAR T-cells in solid tumours demonstrate that they are safe, show promising evidence of clinical activity in very poor prognosis tumours but require further optimisation. TCR-engineered T-cells and other adoptive cell therapy products have shown preliminary safety and evidence of clinical activity in solid tumours. However, none of these have yet received approvals. It is critical to understand the biology and improve efficacy for the rational evaluation of CAR T-cells and other adoptive cell therapies.

6. CAR T-cells in lymphoma

6.1. Paediatric B-NHL

Paediatric mature B-NHL and BCP-ALL are distinct diseases. Most paediatric B-NHLs are Burkitt lymphomas, which are rare and have an excellent prognosis with an RFS of 94% and overall survival (OS) of 95% for patients with high-risk disease and an RFS of 97–98% and >98% OS for standard-risk disease [104]. The acute toxicity of therapy is substantial but there are few expected long-term side effects [103]. Relapses are very rare, with just 50–70 paediatric patients per year, in North America and Europe [1]. The relapsed disease has poor chemosensitivity and a very poor prognosis; however, there are many potential medicinal products for B-NHL, and in view of the very small population, randomised studies are not feasible, and prioritisation is needed. At the ACCELERATE and EMA Paediatric Strategy Forum for mature B cell malignancies based on the mechanism of action and disease specificity, B-cell antigen-targeting CAR T-cells, T-cell engagers and ADC were prioritised for development [1]. Resulting from that Forum, a global academic-led early phase clinical trial to rapidly assess multiple novel agents in paediatric patients with relapsed and refractory B-NHL (GloBNHL) was designed. There are currently three cohorts, bispecific T-cell engager, ADC with standard chemotherapy and CAR T-cells or HSCT. The trial aims to recruit 210 patients over seven years and an efficient Bayesian design will be able to evaluate multiple agents rejecting those that offer no advantage with as few children exposed to an ineffective agent as possible. The trial illustrates the challenge for initial funding of an academic-sponsored, industry-supported platform trial, using the non-frequentist methodology but offers potential solutions for prioritisation strategies.

CD19 CAR T-cells have produced durable responses in adult B NHL [104–106]. Low tumour burden may correlate with a response as in BCP-ALL, but the relationship between CD19 CAR T-cell expansion and response is less than with B- ALL, and durable responses in B-NHL do not require long term persistence of functional CAR T-cells, at least detectable in peripheral blood [104–107]. Many trials, including combination studies in the adult population, are ongoing or upcoming.

Based on biology, the best targets for paediatric B-NHL are the B-lineage markers CD19, CD20 and CD22, and combinatorial targeting may be superior. Pre-clinical studies confirm that BCMA is probably not an optimal target because it is only expressed in late memory B cells committed to plasma cell differentiation and has very limited expression on paediatric Burkitt and DLBCL that arise from earlier stage B cell differentiation. CD19 CAR T-cells have been demonstrated to elicit some activity against paediatric Burkitt lymphoma [108].

The BIANCA (C2202) Novartis Study is a phase II study of tisagenlecleucel in paediatric patients with relapsed/refractory B-NHL [109]. Apheresis, manufacturing and bridging were found feasible, and primary analysis is planned in late 2021 [41]. Outside the trial, in contrast to BCP-ALL and also adult DLBCL, the general picture of early experience with CAR T-cells in relapsed/refractory Burkitt lymphoma is of no response or early partial response followed by rapid progression in a majority of patients. The key questions for future development are (i) what are the mechanisms of resistance, and how could they be overcome? (ii) what are the best targets, and is dual or triple targeting required to optimise therapeutic responses? (iii) what are the comparative benefits of CAR T-cells versus T-cell engagers and ADCs? and (iv) should subsequent HSCT (essentially autologous in this setting) be utilised to consolidate responses?

There is a major difference between BCP-ALL and B-NHL in terms of unmet needs and patient numbers. The number of eligible patients for CAR T-cell studies in paediatric B-NHL is very small, and it is thus not feasible for all CAR T-cell products, even if they are biologically relevant, to be evaluated. Further development requires the results of ongoing paediatric clinical trials (BIANCA [109], ZUMA-4 [41] and JCAR [110] studies), in addition to early and thoughtful prioritisation which is essential in view of limited patient numbers.

6.2. Hodgkin and anaplastic large cell lymphoma

Both Hodgkin lymphoma and anaplastic large cell lymphoma (ALCL) have outstanding OS rates (>95%), although 5y EFS is approximately 75% in ALCL. Many patients with relapsed disease can be salvaged, with targeted therapies, including anti-CD30 immunotoxins, anti-PD1 checkpoint inhibitors, and ALK inhibitors for ALCL. The early activity of CD30 CAR T-cells has been also reported (60% CR; 1-year PFS 36%) [111]. The optimal approach looking forward is the prioritisation of agents with joint paediatric and adult trials as the disease biology is the same. Consideration should be given to evaluate CD30 CAR T-cells in front-line therapy for Hodgkin and other CD30-expressing lymphomas and combination with immune checkpoint anti-PD1 inhibitors due to >90% over-expression of PDL1 and the potential for synergistic activity.

7. Allogeneic therapies and CAR NK-cells

There is increasing evidence that allogeneic CAR T-cells or CAR NK-cells obtained from healthy donors may have roles in childhood malignancies, but the relative utility of these approaches varies by indication. The use of allogeneic CAR T-cells from unrelated, unmatched donors relies on gene-editing to remove TCR genes along with genes that allow at least temporary protection from rejection by host T cells [112]. The main benefit of allogeneic CAR T-cells is their easy ‘off-the-shelf’ availability that may also provide a bridge to other therapies, as their lack of persistence is the main challenge for use as stand-alone agents. Allogeneic CAR T-cells could also overcome challenges in an inability to apheresis and/or manufacture CAR T-cells in patients in whom autologous products are not feasible. Currently, the utility of allogeneic CAR T-cell therapy is relatively less in BCP-ALL, given the success to date of autologous products, but could have an increased role in the treatment of T-ALL, B-NHL, AML and solid tumours.

Cytokine-induced memory-like CAR NK-cells is an alternative approach. CAR NK-cells derive from the innate immune system and recognise targets via CAR with contributions by a complex array of activating and inhibitory receptors [113–116]. There is no relevant risk of graft-versus-host disease given the lack of TCRs on NK cells. Adoptively transferred NK cells have short persistence and cannot establish immune memory, although they have induced potent anti-cancer responses with multi-dose infusions. Investigators at the MD Anderson Cancer Center recently demonstrated that CAR NK-cells derived from cord blood achieved a CR rate of 64% in adults with relapsed/refractory CD-19-positive NHL and chronic lymphocytic leukaemia. In this study, the infused CAR NK-cells expanded and persisted at low levels for at least a year [115]. The main benefit from CAR NK-cells is their ‘off-the-shelf’ availability (overcoming the issues with autologous CAR T-cells of limited access and a complex supply chain) and multiple mechanisms of tumour recognition beyond the CAR alone. CAR NK-cells could have potential roles in AML, T-ALL, lymphomas and solid tumours. Combination approaches with cytokine stimulation or adjunctive therapy to prolong CAR NK-cell survival and persistence *in vivo* are being investigated [115].

8. Products discussed at the Forum

Ten autologous CAR T-cell products, 5 allogeneic CAR T-cell, 1 TCR T-cell, 1 TRuC T-cell and 1 CAR NK-cell product were discussed at the Forum (Table 1). Amongst these, there are currently 5 published paediatric investigation plans (PIPs) agreed with the EMA PDCO for CAR T-cells as of May 2021 (Table 2), all of which target CD19 for BCP-ALL and/or B-NHL.

9. Discussion (Box 1)

9.1. Discussion and conclusions

9.1.1. Patient advocates perspective: Patient advocates believed that the conundrum of how to optimally advance CAR T-cell therapies for paediatric patients with BCP-ALL for relapsed/refractory disease will require additional tightly coordinated regular and ongoing discussion among clinicians. CAR T-cell or other adoptive cell therapeutic options for

solid tumours, AML, and T-ALL are insufficient at this time, and research should be directed to these unmet therapeutic needs. To help address these and other complex clinical strategies, patient advocates urge the use of innovative trial designs, including first-in-child trials when appropriate and enrolling children in adult trials, as well as adaptive designs. Advocates were concerned about potential low enrolment in randomised trials for cell therapies because families may perceive these novel therapies as superior to HSCT or other options. There are concerns that studies may fail to successfully accrue or that evidence will be insufficiently robust as a result, and there are opportunity costs in terms of the time required to undertake them. Patient advocates were also very concerned about the high cost of cell therapy products, associated reimbursement uncertainties, availability and equitable global access. Cost, location and manufacturing time affect patients' access to these novel therapies. While these challenges are outside the purview of clinical research, their solution will require partnerships among researchers and other stakeholders, including pharma companies, funding and health technology assessment (HTA) agencies and advocates. Such partnerships will also be necessary through Paediatric Strategy Forums to help determine which, among a large number of products in development, should be evaluated in rare paediatric cancers. Patient advocates highlighted that, apart from regulatory obligations, long-term toxicities of cell therapies have not been systematically studied [117] and urged industry and academia to establish a registry to document these going forward. They also continued to urge the industry to engage survivors and patient advocates early in drug development, trial design, treatment implementation and patient follow-up to improve the chances that novel scientific insights can offer patients the best valid treatment options.

9.2. Disease-specific discussion

9.2.1. BCP-ALL: The international academic experts, representatives from industry and patient advocates all believed that the role of CAR T-cells in very early and early relapse in a landscape of chemotherapy (+/- blinatumomab/inotuzumab ozogamicin) and HSCT require elucidation. Identifying those patients in whom CAR T-cell therapy can replace HSCT and those patients in whom it may effectively bridge to HSCT is a **very high priority** and requires a randomised study. The TRANSCEND PEDAL study (lisocabtagene maralucel) [110] and the KTE-X19 study (brexucabtagene autoleucel) [41] are evaluating CD19-targeting CAR T-cells, in paediatric patients with relapsed/refractory BCP-ALL and include patients with first marrow relapse. However, more trials need to be initiated in this setting given demonstrated success in second or greater relapse. The results of the CASSIOPEIA trial for patients who are newly-diagnosed with high-risk persistent MRD [40] may clarify the role of CAR T-cells in this scenario, although it is unclear if these results will alter the standard of care. Although not yet proven, dual targeting CD19 and CD22 is the currently most advanced approach for hypothetical prevention of antigen escape. However, the benefits of this approach need to be established. Therefore, additional products targeting CD19 alone should only be developed in paediatrics if the product has a substantial benefit (demonstrated from adult and/or preliminary paediatric studies) in terms of efficacy, persistence and/or toxicity compared to existing products. Development of CAR T-cells with improved persistence and more favourable immunologic properties (e.g. human or humanised CAR constructs, use of advanced culture conditions and manufacturing technologies enriching for T cells with superior fitness and longevity) is another priority but

does not address the problem of antigen escape. In addition, strategies must evolve to ensure appropriate access for children who could potentially benefit from this therapy.

9.2.2. AML and T-ALL: CAR T-cells are early in development for AML and T-ALL. CD33 and CD123 appear to be current optimal targets in AML and CD7 in T-ALL. The results of ongoing or planned first-in-child studies of autologous products are required to facilitate further understanding. Allogeneic CAR T-cells, TCR T-cells, NK and CAR NK-cells, and antibody-based immunotherapies may also be beneficial in these heavily pre-treated populations in which autologous T cell apheresis may be suboptimal and require further study.

9.2.3. Solid Tumours: Promising early results have been reported, particularly with GD2 targeting, and there is substantial potential to address a significant unmet clinical need. It is critical to understand the biology of the solid tumour microenvironment with its barriers against T cells and improve targeting efficacy and selective targeting to rationally evaluate CAR T-cells in solid tumours. Alternatives such as CAR NK-cells and engineered T cells that use the entire TCR must also be considered. Cell therapies may require a combination with other immuno-oncology agents to be active against solid cancers.

9.2.4. Lymphoma: B-NHL: There are many differences between paediatric B-NHL and BCP-ALL. Prioritisation is needed, and randomised studies are not feasible to evaluate CAR T-cells in Burkitt lymphoma in view of the very small patient numbers. The early experience with CAR T-cells in this situation comprises, mostly, no responses or early relapses. The BIANCA phase II study is expected to demonstrate the feasibility and initial efficacy of CD19 CAR T-cells in paediatric relapsed/refractory B-NHL. Mechanisms of resistance must be understood to inform further development and combination trials with alternative targets (CD20 or CD22) should also be explored as a priority to improve efficacy in this population.

9.2.5. Hodgkin lymphoma and ALCL: The optimal approach for product development of CD30 CAR T-cells in Hodgkin disease and ALCL is joint paediatric and adult trials. CAR T-cells targeting other antigens may also be relevant for these patients.

9.3. General discussion

9.3.1. Combinatorial regimens: In many paediatric malignancies, a combinatorial approach with cell therapies may be warranted, including multi-antigen targeting. Other combinations may include cell therapies in conjunction with (i) immune checkpoint inhibitors; (ii) anti-tumour vaccines; (iii) cytokine stimulation; (iv) therapies to up-regulate target antigen expression; (v) modulators of the tumour micro-environment; (vi) chemotherapy; (vii) kinase or other small molecule inhibitors to prevent CAR T-cell tonic signalling and subsequent exhaustion; (viii) bispecific, trispecific, quad specific T-cell engager antibodies to extend the target spectrum. Decisions of the appropriate combinations should be based on the biology of the individual cancers, sufficient pre-clinical and clinical data and identification of the contribution of each component based on an appropriate study design.

9.3.2. Access for children to cell therapies: Currently, many children who could potentially benefit from such therapy, either through the approved product or a clinical trial, do not have access to this innovation. Strategies must evolve to ensure appropriate access.

9.3.3. Decentralised academic and non-academic manufacturing: The challenges of access to CAR T-cell products in Europe was highlighted in 2017, when it was apparent that <10% of CAR T-cell clinical trials worldwide were conducted in Europe [8]. To facilitate greater access for children and adolescents with malignancy to CAR T-cell products, a decentralised model of manufacture potentially has advantages, including improved logistics with manufacturing closer to the patient. Semi-automated closed-system manufacturing devices now allow in-house manufacturing of standardised products by academic institutions upon product-specific licences. Academic manufacturing, with qualification of both at the manufacturing facility and the sites, could provide access to products within investigator-initiated multicentre trials for rare conditions where academia has the scientific lead. Realistic requirements for decentralised manufacturing that is now occurring in terms of quality control assays and release criteria need to be agreed upon by regulators, industry and academia.

9.3.4. CAR T-cell trial design: The regulatory/ethical aspects of executing first-in-human CAR T-cell studies in children and adults need to be considered. The general dogma is that first-in-human dose-finding studies should be carried out in adults first and then in children. However, there is a lack of precedence in study designs that take into account that age is a continuum, and in general, children tolerate T-cell therapies much better than adults and there are, for instance, more BCP-ALL cases in childhood than in adulthood. Therefore, new clinical study designs are needed for early phase clinical studies enrolling both children and adults; data from either group could be used to inform the other, as long as subentities of disease are comparable. For paediatric studies of CAR T-cells, the requirement to demonstrate safety in adults and teenagers first before enrolling younger patients is not logical because CAR T-cell therapy, to date, has largely induced fewer or more manageable toxicities in children than in adults. The use of novel trial designs should be used to reduce the required number of patients and evaluate new cell therapies most efficiently.

Generally, CAR T-cell products being developed for adults are not a route for meaningful paediatric studies. Because of, sometimes, distinct biologies and cells of origin of paediatric and adult malignancies, only a limited overlap exists for cancers and antigens of interest. However, the inclusion of adolescents in adult programs, if scientifically justified, is highly encouraged to improve the efficiency of trial enrolment and maximise access to novel therapies for older paediatric patients.

An additional challenge is ensuring access to clinical trials, especially where the disease is very rare, and there is a very small trial population. Referral of patients, including cross-border referrals, should be strongly encouraged. Patient advocates, academia, and industry should work together to ensure adequate cost coverage in these situations. The model where studies determining the dose and toxicity profile are carried out in a limited number of centres, but then there is a rapid expansion of sites, would allow greater access to children to innovative therapies and is strongly supported.

9.3.5. CAR T-cells, ADCs and T-cell engaging products: CAR T-cell products for specific diseases based on relevant antigen expression will be directly competitive with both ADCs and T-cell engaging products directed against the same target. Therefore, a key question is to determine the comparative benefits of CAR T-cells versus T-cell engagers and ADCs in specific situations. The ease with which the T-cell engagers and ADCs can be administered to patients, in comparison to the challenges required to administer CAR T-cell products, will make them highly attractive if they have reasonably comparable activity. The issue of competition with T-cell-engaging products and ADCs applies perhaps even more in the solid tumour setting because in the B-cell directed leukaemia/lymphoma setting, CAR T-cell persistence may be feasible. In the solid tumour setting, persistence has not yet been achieved, and multiple infusions seem likely to be required for a sustained effect. Conceptual advantages of gene-engineered T-cells over engagers or ADCs in solid tumours, such as improved trafficking to metastatic sites of disease and infiltration into tumours, have not yet been demonstrated in clinical therapy. There are ADCs and/or T-cell engaging products for many of the solid tumour antigens also targeted by CAR T-cells, including HER2 and B7-H3/CD276 that are under early clinical evaluation. As shown in acute childhood leukaemias for blinatumomab and inotuzumab and adult solid tumours (trastuzumab deruxtecan for HER2-positive advanced-stage breast cancer [118] and trastuzumab deruxtecan against HER2-positive gastric cancer [119]), response rates for these agents, in general, can be substantial. Decisions on prioritising and designing clinical trials for CAR T-cell products targeting these antigens will need to take into consideration plans for and results from clinical trials for the comparable ‘off-the-shelf’ ADC and T-cell engaging products.

9.3.6. Development plan for cell therapies: Very early planning of the development pathway is critical for optimal efficiency. Regulators should be involved from the inception agreeing with the design of clinical trials of innovative medicines for which regulatory approval may ultimately be sought. Also, early academia-multi-company discussion and potential collaboration may be beneficial. By aligning academic, regulatory and HTA (in the EU) requirements from the inception of a clinical trial, drug development will be accelerated, the patients with the greatest need will be prioritised and evidence for scientific and regulatory purposes will be generated. Trial design (randomised versus non-randomised), identification of appropriate ‘control’ populations (historical versus contemporaneous) and comparisons with the standard of care are critical issues. It is important to define the final target population, for example, very high-risk front-line disease and design studies that allow for early decision making of development plans to continue or not, while overall ensuring clinically relevant endpoints and robust comparative data are generated to also inform HTAs. The inclusion of front-line trials in PIPs may also help align scientific, regulatory and HTA requirements; however, a focussed and sequential approach must be adopted to ensure an appropriately sized population is available for each trial. The challenge occurs when trials are required to determine the standard of care after registrational studies. Treatment optimisation is essential for paediatric cancers and is long-established, cooperative international clinical studies. An additional challenge is establishing the contribution of components for regulatory support when seeking to develop CAR T-cell in combination with other agents. There needs to be a dynamic process for prioritisation.

The current approach for increasing trans-Atlantic regulatory (EMA and FDA) alignment, for example, the simultaneous submission of PIPs and iPSPs and the published common commentary document was applauded [120–122]. Harmonisation of different countries within the European Union would be of great benefit to both academia and industry.

9.3.7. Development of cell therapy products by academia: In general, the industry should be encouraged to partner with academia to help support rare disease indications in paediatric oncology. However, there are scenarios when it is anticipated that the industry will not be willing to develop a CAR T-cell or TCR T-cell product, for example, for very small paediatric populations, and where trials are required to determine the standard of care, but these are not registrational studies. Unlike small molecules and protein therapeutics where the timing of first-in-child trials is generally determined by industry, first-in-child trials of cell therapies can be and have been launched without private sector investments. The model for drug development for cancer cell therapy in paediatrics could involve a ‘later stage handoff’ to industry after the conduction of first studies in the academic arena. This approach could enable more rapid/efficient yet robust testing of the most promising products, targets and combinations. The optimal position is one in which there is a partnership with the industry through innovative collaborative platforms. A flexible model is needed, as ‘one size does not fit all’. Co-development and funding are other options. An academic development model for cell therapy products relies on academic manufacturing of uniform products by good manufacturing practice standards upon local licences, along with potential distribution to additional academic trial sites (decentralised academic manufacturing). In this model, the academic manufacturing centre would ensure quality to good manufacturing practice levels and take responsibility for all regulatory related actions. However, this model would require substantial financial investments by the government, foundations and academic medical centres, which have some track record of experience and success and requires further discussions moving forward. It is also critical that access for all children with relevant diseases is available when development is finalised and drugs are validated.

10. Conclusion

The development of CAR T-cell therapy and adoptive cellular therapies is very rapidly progressing and evolving and hold much promise. This is a field where first-in-human CAR T-cell studies have occurred in children in the past and this needs to continue in the future. For the benefit of children and adolescents with cancer, close partnerships between academia and industry are essential. The relative utility of different approaches (autologous and allogeneic CAR T-cell, TCR T-cell, TRuC T-cells, CAR NK-cell ADCs and T-cell engaging products) varies by indication. Therefore, intense discussion between academia and industry, with early regulatory advice, will be required to obtain a consensus. As these are very expensive therapies, aligning strategic, scientific, regulatory and funding requirements from the inception of a clinical trial is of critical importance. It is also imperative that strategies evolve to ensure appropriate ease of access for children who need and could potentially benefit from these therapies.

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Box 1.**Text box of key conclusions of the Paediatric Strategy Forum.**

- Chimeric antigen receptor (CAR) T-cell therapy has substantial promise for children and adolescents with cancer, but there are biological, clinical and manufacturing challenges, as well as significant challenges to access for many children
- Anti-CD19 CAR T-cells produce high response rates in relapsed/refractory B-cell ALL, although the effect is of limited duration in half of the patients
- Targeting multiple antigens to prevent antigen escape and the use of humanised products to improve persistence could optimise therapy for B-cell ALL
- CAR T-cell treatment could potentially benefit children and adolescents with very high-risk B-cell ALL who ultimately fail conventional therapies
- Identifying patients with B-cell ALL with very early and early relapse or those with the persistent initial disease in whom CAR T-cell therapy can replace haematopoietic stem cell transplantation and those patients in whom it can be a more effective bridge to haematopoietic stem cell transplantation than current options is a very high priority
- Prioritisation is needed to evaluate CAR T-cells in Burkitt lymphoma
- Combination trials of additional or alternative targets (CD20 or CD22) should be considered in Burkitt lymphoma
- Development of CD30 CAR T-cells in Hodgkin lymphoma and ALCL should be by joint paediatric and adult trials
- CAR T-cells are early in development for AML and T-ALL now in phase 1 clinical trial testing
- CD33 and CD123 appear to be good targets in AML and CD7 in T-ALL; other targets are under evaluation
- There are promising early results in solid tumours, particularly with GD2 targeting, and there is substantial potential to address a significant unmet clinical need
- Novel CAR T-cell designs, for example, allow T-cell activation only in the presence or absence of an additional marker, and combinations of targets and (immuno)therapies need to be investigated in solid tumours
- The comparative benefits of autologous versus allogeneic CAR T-cells, T-cell receptor-engineered T-cells, TRuCs, NK cells and CAR NK-cell products, T-cell engager antibodies, and antibody-drug conjugates require evaluation

- Global academic collaboration, very early involvement of regulators in studies seeking registration, early academia-multi company-patient advocacy engagement are critical factors
- The model for drug development for cancer cell therapy in paediatrics could involve a 'later stage handoff' to industry after first clinical evaluation in the academic arena
- To facilitate greater access for children and adolescents with malignancy to CAR T-cell products and allow for the performance of academic multicentre studies establishing a new standard of care, a decentralised model of manufacture potentially has advantages
- Continued development is required for many CAR T-cells after the first approved indication. However, currently, there is neither requirement nor any incentives for a company to continue to support innovation in the same patient population once a PIP or iPSP has been completed, or an approved indication obtained
- Patient advocates urged investigators to agree and drive forward key research questions strategically for BCP-ALL and solid tumours, strongly emphasised the need to broaden access to these novel treatments, and a common data registry to be used by both academia and industry to capture important long-term follow-up information.
- Strategies must evolve to ensure appropriate access (including cross-border access) for children who could potentially benefit from these therapies

Table 1

Medicinal products discussed at the Paediatric Strategy Forum.

Class of medicinal product	Product	Target	Company
Autologous CAR T-cell	JCAR017/Ilisocabtagene maraleucel (liso-cel) Breyanzi [®]	CD19	BMS/Celgene
Autologous CAR T-cell	Tisagenlecleucel, Kymriah [®]	CD19	Novartis
Allogenic CAR T-cell	CTX110	CD19	CRISPR Therapeutics
Autologous CAR T-cell	KTE-X19	CD19	Kite Pharma/Gilead
Autologous CAR T-cell	AUTO1	CD19	Autolus
Autologous CAR T-cell	AUTO1/22	CD19 & CD22	Autolus
Allogenic CAR T-cell	UCART22	CD22	Collectis
Autologous CAR T-cell	Syncopation CD22	CD22	Syncopation Life Sciences
TCR-T cell	Letetresgene autoleucel (lete-cel); GSK3377794	NY-ESO-1	GlaxoSmithKline
NK CAR	TAK-007		Takeda
Allogenic CAR T-cell	GC027	CD7	Gracellbio
Allogenic CAR T-cell	UCART 123	CD123	Collectis
T cell receptor fusion construct (TRuC)	Gavocabtagene autoleucel/gavo-cel (formerly TC-210)	Anti-mesothelin.	TCR ² Therapeutics
Allogenic CAR T-cell	CTX130	CD70	CRISPR Therapeutics
Autologous CAR T-cell	TT11	CD30	Tessa Therapeutics
Autologous CAR T-cell	MB-CART2019.1	CD19 & CD20	Miltenyi Biomedicine
Autologous CAR T-cell	MB-CART2219.1	CD19 & CD22	Miltenyi Biomedicine
Autologous CAR T-cell	MB-CART19.1	CD19	Miltenyi Biomedicine

Table 2

Published PIPs agreed for CAR T-cells.

Product	Brexucabtagene Autoleucel - <i>Tecartus</i> KTE-X19 (Kite-Gilead)	Axicabtagene ciloleucel - YESCARTA (Kite-Gilead)	Tisagenlecleucel (Novartis)	Tisagenlecleucel (Novartis)	JCAR017, lisocabtagene maraleucel (Celgene)
PIP	EMA-001862-PIP01-15-M02 Decision No P/0142/2020; date 18/04/2020	EMA-002010-PIP01-16-M02 Decision No P/0132/2020; date 15/04/2020	EMA-001654-PIP02-17-M01 (Decision No P/0323/2019; date 11/09/2019)	EMA-001654-PIP01-14-M03 (Decision No P/0008/2019; date 03/01/2019)	EMA-001995-PIP01-16-M02 (Decision No P/0198/2019; date 12/06/2019)
MoA	CD19	CD19	CD19	CD19	CD19
Condition	Acute lymphoblastic leukaemia	Mature B-cell neoplasms	Treatment of mature B-cell neoplasms	Treatment of B cell ALL/NHL	Treatment of ALL/NHL
PIP Indication	Paediatric patients with r/r B-ALL	Paediatric patients with r/r B-NHL	Treatment of paediatric patients with CD19+ relapsed or refractory NHL	Treatment of CD19+ refractory/relapsed B-ALL in paediatric patients	Treatment of paediatric patients with CD19+ relapsed or refractory B-ALL/DLBCL, BL or PMLBCL
Waiver	Paediatric population weighing less than 6 kg				
Deferral	Yes. For completion by December 2023 . No published compliance check yet.	Yes. For completion by December 2023 . No published compliance check yet.	Yes. For completion by March 2022. No published compliance check yet.	Yes. For completion by November 2026. No published compliance check yet.	Yes. For completion by November 2023. No published compliance check yet.
Formulation	Dispersion for infusion; intravenous use.				
Clinical	Safety and activity r/r B-ALL or B NHL	Safety and activity r/r B-ALL or B NHL	Safety and activity with CD19+ relapsed or refractory B-NHL	1 Safety and feasibility chemotherapy-resistant or refractory CD19+ leukaemia or lymphoma 2 Safety and activity refractory/relapsed CD19+ B-ALL/NHL 3 Safety and activity refractory/relapse CD19+ B-ALL 4 Efficacy and safety in paediatric patients with <i>de novo</i> high-risk 1–18 years of age B-ALL MRD+.	1 Activity and safety relapsed or refractory CD19+ B-ALL or B-NHL 2 Safety, feasibility and efficacy with relapsed or refractory CD19+ B-ALL